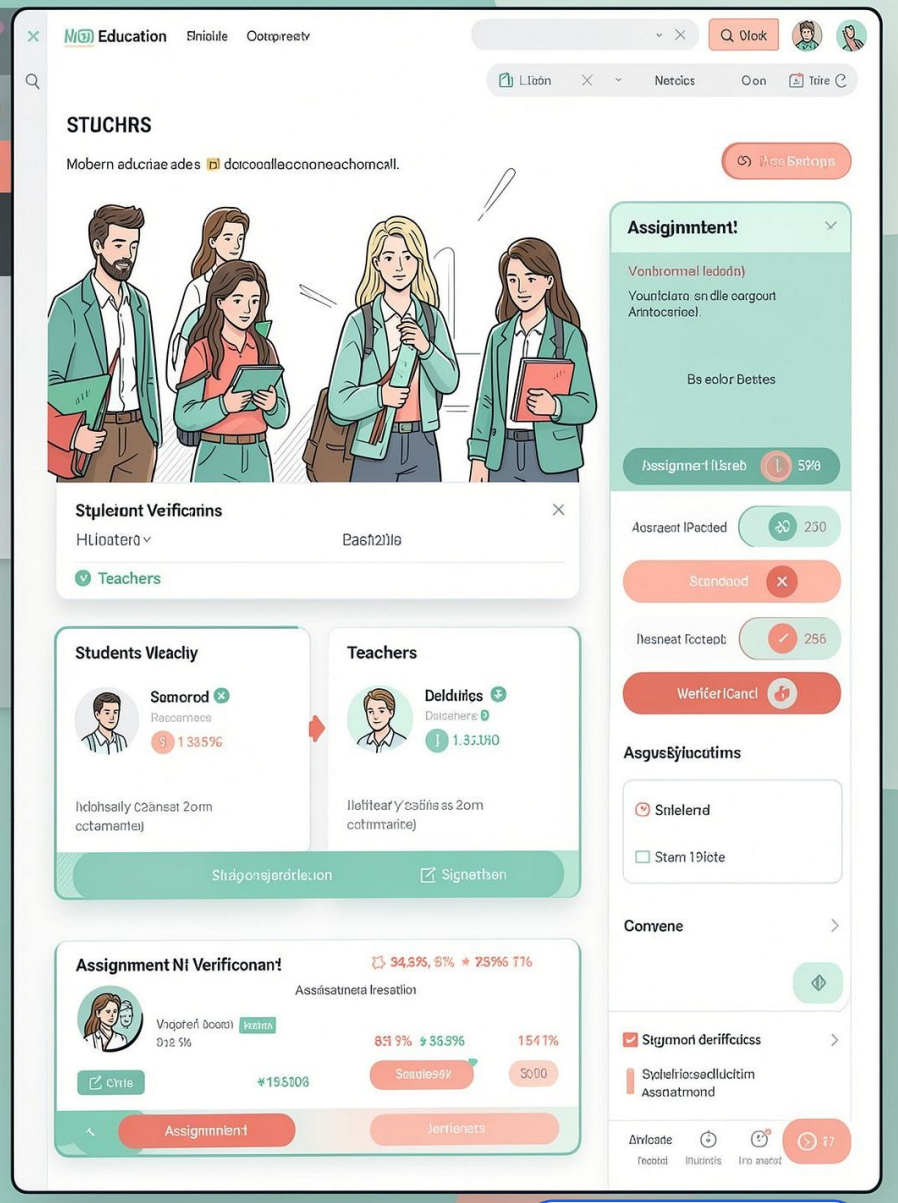
Smart Student Portal

Assignment Verification & Smart Academic Management System

MERN Stack Project

“Submit. Verify. Improve.”



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What is the Smart Student Portal?

Smart Student Portal is a web-based platform that connects Students, Teachers, and Administrators in one system — with one unique focus: assignment verification before final submission.



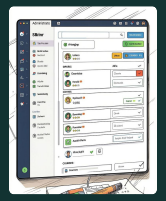
STUDENT

- View courses
- View & submit assignments
- View attendance & fees
- Receive notifications



TEACHER

- Create assignments
- Define exact requirements
- Mark attendance
- Review submissions & give feedback



ADMIN

- Manage students & teachers
- Manage courses & batches

MAIN FEATURE ⭐ Assignment Verification System (visually larger and prioritized)

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What Problem Does Our System Solve?

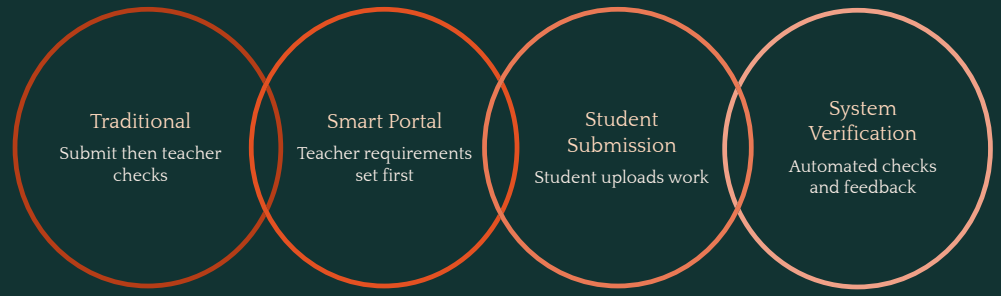
Traditional Flow

Teacher assigns → Student completes → Student submits → Teacher manually checks

- ❌ Student may miss required topics
- ❌ Required features might be missing
- ❌ Incomplete submissions accepted
- ❌ Teacher spends time on basic checks
- ❌ Absent students miss assignments and topics

Our Solution

Verify the assignment before final submission.



SMART PORTAL: Requirements → Submission → System Verification → Teacher Review

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How Assignment Verification Works



Create Define Complete Submit Verify

1

Step 1 — Create

Teacher adds assignment metadata and deadline.

2

Step 2 — Define Requirements

Teacher lists exact required features and files (example list below).

3

Step 3 — Complete & Submit

Student builds solution and uploads code/files.

4

Step 4 — System Verification

Automated checks compare submission to requirements.

5

Step 5 — Outcome

Accepted → Teacher notified; Missing → Blocked with detailed feedback for resubmission.

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Real Example: JavaScript Assignment

Teacher Requirements

Requirement	Required
DOM Manipulation	✓
Event Handling	✓
Add Button	✓
Delete Button	✓
Local Storage	✓
API Integration	✓

Student Submission Check (Dashboard View)



DOM Manipulation

✓ PASSED



Event Handling

✓ PASSED



Add / Delete Button

✓ PASSED



Local Storage

❌ MISSING



API Integration

✓ PASSED

RESULT: SUBMISSION NEEDS CORRECTION — Missing: **Local Storage**

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After Requirement Verification: Automated Testing

Example Test Suite (visual report)

Test	Status
Button Click Test	✅ Passed
Form Submission Test	✅ Passed
API Data Loading	❌ Failed
Required Page Present	✅ Passed
Required File Check	✅ Passed
Basic Error Check	⚠️ Issue Found

Verification Report (Summary)

Note: Automated testing assists the teacher. It does not replace teacher evaluation and cannot perfectly grade complex or subjective code tasks.

Requirements

6 / 6

Tests Passed

4 / 5

Status

Teacher Review Required

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Smart Notification for Absent Students



TODAY'S CLASS

Total Students	50
Present	30
Absent	20

Teacher Topic: **JavaScript DOM Events**

Assigned: **DOM Events Practice**

System auto-identifies absent students and sends a notification card

Notification Card (example)

🔔 **You Missed Today's Class**

Topic: JavaScript DOM Events
Assignment: DOM Events Practice
Message: “Please review the topic and complete the assignment before the deadline.”

Benefit: Keeps absent students connected to class learning.

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Complete Smart Student Portal Workflow



Side flow: Attendance → Absent student identified → Automatic notification → Missed topic + assignment

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Technology Stack & System Architecture

Frontend

- React.js
- Tailwind CSS

Backend

- Node.js + Express.js (REST APIs)
- Assignment Verification Logic in Node

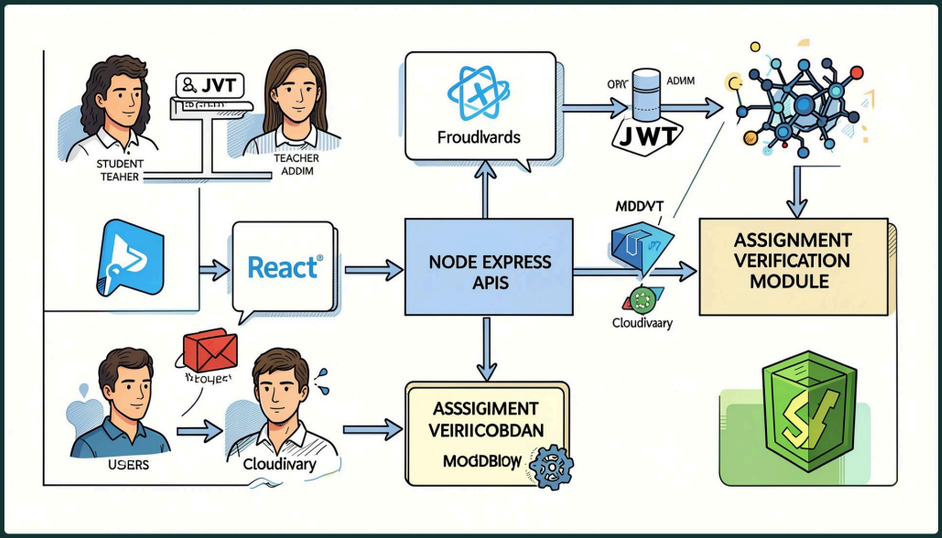
Database

- MongoDB with Mongoose

Supporting

- JWT + bcrypt (auth)
- Multer + Cloudinary (file upload)
- Git & GitHub (version control)
- JavaScript as main language

Core = MERN Stack + JavaScript. No Python/Java/C++ shown.



Student / Teacher / Admin → React → Express + Node → Verification Logic → MongoDB

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Why Is Our Project Different?

1. Assignment Requirement Verification

Teachers define exact, testable requirements.

2. Pre-Submission Checking

Students discover missing items before final submit.

3. Automated Functionality Checks

Predefined tests verify basic behaviors (buttons, API load, form).

4. Verification Report

Structured report guides teacher review and student correction.

5. Absent Student Notifications

Auto-notifies absent students with topic + assignment.

Future Scope

- AI-assisted code analysis (research stage)
- Advanced automated testing
- Plagiarism detection
- Student performance analytics
- Mobile app, Email/SMS notifications
- Multi-institute support

Final note: The goal is not to replace the teacher. It reduces repetitive checking, helps students find missing requirements, and keeps students connected.

“Submit. Verify. Improve.”

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